1 Triangle books with leaves rooted at page vertices

For a finite graph H and $0 \le p < 1$, put

$$\Phi_H(p) := (1-p)^{v(H)} \chi_H\left(\frac{1}{1-p}\right),$$

and use its polynomial continuation at $p = 1$. We prove a positive theorem for triangle books whose pendant leaves are distributed among the page vertices. This is complementary to the previously proved two-root theorem, in which leaves are attached to the two ends of the common spine edge.

Throughout, $(\Omega, \mathcal{A}, \mu)$ is an arbitrary probability space and $W : \Omega^2 \to [0,1]$ is a symmetric jointly measurable graphon. Write

$$p = t(K_2, W) = \int_{\Omega^2} W(x,y)\, d\mu(x)\, d\mu(y), \qquad d(x) = \int_{\Omega} W(x,y)\, d\mu(y).$$

2 The graph family and its exact target

Definition 2.1 (Page-rooted leaf book)**.** *Let $m \ge 1$ and let $\mathbf{k} = (k_1, \dots, k_m)$ be a vector of nonnegative integers. The graph $P_{m;\mathbf{k}}$ has two adjacent* spine vertices *u, v, page vertices $z_1, \dots, z_m$, and k_i private pendant leaves adjacent only to z_i. Each page vertex is adjacent to both spine vertices, and there are no other edges.*

Put

$$n := k_1 + \cdots + k_m.$$

Then

$$v(P_{m;\mathbf{k}}) = m + n + 2, \qquad e(P_{m;\mathbf{k}}) = 2m + n + 1.$$

Lemma 2.2 (Chromatic data)**.** *For every $m \ge 1$ and every $\mathbf{k} \in \mathbb{N}^m$,*

$$\chi(P_{m;\mathbf{k}}) = 3$$

and

$$\chi_{P_{m;\mathbf{k}}}(x) = x(x-1)^{n+1}(x-2)^m. \tag{1}$$

Consequently, for every $p \in [0,1]$,

$$\Phi_{P_{m;\mathbf{k}}}(p) = p^{n+1}(2p-1)^m. \tag{2}$$

Proof. Choose the two distinct colours on the ordered spine in $x(x-1)$ ways. Each page vertex then has $x-2$ choices, independently of the other pages. After a page colour has been chosen, each of its private leaves has $x-1$ choices. This proves (1). The graph contains a triangle and the same construction gives a 3-colouring, so its chromatic number is 3.

For $q = 1 - p > 0$, substitution gives

$$\begin{aligned} \Phi_{P_{m;\mathbf{k}}}(p) &= q^{m+n+2} \frac{1}{q} \left(\frac{p}{q}\right)^{n+1} \left(\frac{2p-1}{q}\right)^m \\ &= p^{n+1}(2p-1)^m. \end{aligned}$$

The last expression is a polynomial in p and therefore also gives the continued value 1 at $p = 1$. □

3 A weighted rooted-triangle inequality for real exponents

Define

$$\tau(x) := \int_{\Omega^2} W(x,y)W(x,z)W(y,z)\,d\mu(y)\,d\mu(z)$$

and, for every real $s \geq 0$,

$$M_s := \int_{\Omega} d(x)^s\,d\mu(x),$$

with the convention $d(x)^0 = 1$, including when $d(x) = 0$.

Lemma 3.1 (Weighted rooted triangle)**.** *If $p > 0$, then for every real $h \geq 0$,*

$$\int_{\Omega} d(x)^h \tau(x)\,d\mu(x) \geq \frac{2p-1}{p} M_{h+2}. \tag{3}$$

Proof. For a bounded measurable kernel K, write

$$(T_K f)(x) := \int_{\Omega} K(x,y)f(y)\,d\mu(y), \qquad \langle f, g\rangle := \int_{\Omega} f(x)g(x)\,d\mu(x).$$

Set

$$I_h := \int d^h\tau, \qquad J_h := \langle d^h, T_W d\rangle.$$

The elementary inequality $ab \geq a + b - 1$ on $[0,1]^2$, applied to $W(x,y)$ and $W(x,z)$ and then multiplied by the nonnegative factor $d(x)^h W(y,z)$, gives after Tonelli and symmetry

$$I_h \geq 2J_h - pM_h. \tag{4}$$

Put $U = 1 - W$, $u = T_U\mathbf{1} = 1 - d$, and $q = 1 - p$. Since

$$T_W d = d - q\mathbf{1} + T_U u,$$

we have

$$J_h = M_{h+1} - qM_h + \langle d^h, T_U u\rangle. \tag{5}$$

Symmetry of U gives the exact identity

$$\begin{aligned} &2\left(\langle d^h, T_U u\rangle - \int_{\Omega} d(x)^h u(x)^2\,d\mu(x)\right) \\ &\quad = \int_{\Omega^2} U(x,y)\big(u(y) - u(x)\big)\big(d(x)^h - d(y)^h\big)\,d\mu(x)\,d\mu(y). \end{aligned}$$

Because $d = 1 - u$ and $s \mapsto s^h$ is nondecreasing on $[0,1]$, the integrand on the right is nonnegative. Therefore

$$\langle d^h, T_U u\rangle \geq \int d^h(1-d)^2 = M_h - 2M_{h+1} + M_{h+2}.$$

Substituting this into (5), and then into (4), yields

$$I_h \geq 2M_{h+2} - 2M_{h+1} + pM_h.$$

Finally,

$$\begin{aligned} &2M_{h+2} - 2M_{h+1} + pM_h - \frac{2p-1}{p} M_{h+2} \\ &\quad = \frac{1}{p}\int_{\Omega} d(x)^h (d(x) - p)^2\,d\mu(x) \geq 0. \end{aligned}$$

This proves (3). Notice that no integrality of h was used; this is the point needed below. □

4 Compression of the page-rooted moments

For $x, y \in \Omega$ and real $s \geq 0$, define

$$H_s(x,y) := \int_\Omega W(x,z)W(y,z)d(z)^s \, d\mu(z).$$

Lemma 4.1 (Hölder compression)**.** *Let $k_1, \ldots, k_m \geq 0$ be real numbers and put*

$$\alpha := \frac{k_1 + \cdots + k_m}{m}.$$

Then, pointwise for all $x, y \in \Omega$,

$$\prod_{i=1}^m H_{k_i}(x,y) \geq H_\alpha(x,y)^m. \tag{6}$$

Proof. Apply generalized Hölder inequality to the m functions

$$z \longmapsto \big(W(x,z)W(y,z)d(z)^{k_i}\big)^{1/m}.$$

Their pointwise product is $W(x,z)W(y,z)d(z)^\alpha$. Thus

$$H_\alpha(x,y) \leq \prod_{i=1}^m H_{k_i}(x,y)^{1/m},$$

which is equivalent to (6). The convention $d^0 = 1$ handles zero exponents, and no quotient is used. □

5 The page-rooted leaf theorem

Theorem 5.1 (Leaves on the pages of a triangle book)**.** *Let $m \geq 1$ and $\mathbf{k} = (k_1, \ldots, k_m) \in \mathbb{N}^m$. For every graphon W of edge density $p \in [1/2, 1]$,*

$$\boxed{t(P_{m;\mathbf{k}}, W) \geq p^{n+1}(2p-1)^m = \Phi_{P_{m;\mathbf{k}}}(p).}$$

This is the full interval required by $\chi(P_{m;\mathbf{k}}) = 3$.

Proof. Assume first that $1/2 \leq p < 1$; in particular $p > 0$. Integrating out every private leaf and then every page vertex gives the exact identity

$$t(P_{m;\mathbf{k}}, W) = \int_{\Omega^2} W(x,y) \prod_{i=1}^m H_{k_i}(x,y) \, d\mu(x) \, d\mu(y). \tag{7}$$

All integrands are nonnegative and bounded, so Tonelli justifies this factorization on an arbitrary probability space.

Put $\alpha = n/m$. By Lemma 4.1,

$$t(P_{m;\mathbf{k}}, W) \geq \int_{\Omega^2} W(x,y) H_\alpha(x,y)^m \, d\mu(x) \, d\mu(y).$$

Use the edge-biased probability measure

$$d\rho(x,y) := \frac{W(x,y)}{p}\, d\mu(x)\, d\mu(y).$$

Jensen's inequality for the convex function $s \mapsto s^m$ gives

$$\begin{aligned} t(P_{m;\mathbf{k}}, W) &\geq p\big(\mathbb{E}_\rho H_\alpha\big)^m \\ &= p^{1-m}\left(\int_{\Omega^2} W(x,y) H_\alpha(x,y)\, d\mu(x)\, d\mu(y)\right)^m. \end{aligned} \tag{8}$$

Expanding H_α and applying Tonelli once more gives

$$\int_{\Omega^2} W(x,y) H_\alpha(x,y)\, d\mu(x)\, d\mu(y) = \int_\Omega d(z)^\alpha \tau(z)\, d\mu(z).$$

Lemma 3.1 at the real exponent $h = \alpha$ and Jensen's inequality for $s \mapsto s^{\alpha+2}$ imply

$$\int d^\alpha \tau \geq \frac{2p-1}{p} M_{\alpha+2} \geq \frac{2p-1}{p} p^{\alpha+2} = (2p-1)p^{\alpha+1}.$$

Substitution into (8), using $m\alpha = n$, yields

$$\begin{aligned} t(P_{m;\mathbf{k}}, W) &\geq p^{1-m}\big((2p-1)p^{\alpha+1}\big)^m \\ &= p^{n+1}(2p-1)^m. \end{aligned}$$

By Lemma 2.2, this is exactly the required target.

If $p = 1$, then $0 \leq W \leq 1$ and $\int W = 1$ imply $W = 1$ almost everywhere. Every finite homomorphism density is then 1, equal to the polynomially continued target. □

6 A conditional cone consequence

For $d(x) > 0$, define the probability measure

$$d\mu_x(y) := \frac{W(x,y)}{d(x)}\, d\mu(y),$$

and regard the same kernel W as a graphon W_x on $(\Omega, \mathcal{A}, \mu_x)$. Its edge density is

$$z_x = t(K_2, W_x) = \frac{\tau(x)}{d(x)^2}.$$

Put $z_x = 0$ when $d(x) = 0$.

Lemma 6.1 (Conditional cone lemma)**.** *Let F be a graph on $N \geq 2$ vertices and put $\phi(z) = \Phi_F(z)$. Suppose that, for some $a \in [0,1)$:*

1. *every graphon V on every probability space, of edge density $z \geq a$, satisfies $t(F,V) \geq \phi(z)$;*
2. *$\phi \geq 0$, $\phi' \geq 0$, and $\phi'' \geq 0$ on $[a,1]$;*
3. *$\phi(a) = 0$.*

Then every graphon W of edge density

$$p \geq \frac{1}{2-a}$$

satisfies

$$t(K_1 \vee F, W) \geq \Phi_{K_1 \vee F}(p).$$

Proof. The endpoint $p = 1$ is immediate, so assume $p < 1$ and put $c = 2 - 1/p$. The density hypothesis is exactly $c \geq a$, while $c \leq 1$. Every graphon V of arbitrary edge density $z \in [0,1]$ satisfies the global tangent bound

$$t(F, V) \geq \phi(c) + \phi'(c)(z - c). \tag{9}$$

Indeed, for $z \geq a$ this follows from the assumed bound and convexity. For $z < a$, the tangent line is nonpositive at a and has nonnegative slope, so it remains nonpositive; then $t(F, V) \geq 0$ proves (9).

Conditioning on the image of the cone vertex gives

$$t(K_1 \vee F, W) = \int_\Omega d(x)^N t(F, W_x)\, d\mu(x).$$

The set $\{d = 0\}$ contributes zero and entails no division. Applying (9) in each nonzero link yields

$$t(K_1 \vee F, W) \geq M_N \phi(c) + \phi'(c) \left(\int d(x)^N z_x \, d\mu(x) - c M_N \right).$$

By Lemma 3.1 with $h = N - 2$,

$$\int d(x)^N z_x \, d\mu(x) = \int d(x)^{N-2} \tau(x)\, d\mu(x) \geq c M_N.$$

Thus the second term is nonnegative. Since $\phi(c) \geq 0$ and $M_N \geq p^N$ by Jensen,

$$t(K_1 \vee F, W) \geq p^N \phi(c).$$

Let $q = 1 - p > 0$. Since $1 - c = q/p$ and $\chi_{K_1 \vee F}(x) = x \chi_F(x - 1)$,

$$p^N \phi(c) = q^N \chi_F\left(\frac{p}{q}\right) = q^{N+1} \chi_{K_1 \vee F}\left(\frac{1}{q}\right) = \Phi_{K_1 \vee F}(p).$$

□

Theorem 6.2 (Cones over page-rooted leaf books)**.** *For every $m \geq 1$ and $\mathbf{k} \in \mathbb{N}^m$, the cone*

$$C_{m;\mathbf{k}} := K_1 \vee P_{m;\mathbf{k}}$$

satisfies, for every graphon of edge density $p \in [2/3, 1]$,

$$\boxed{t(C_{m;\mathbf{k}}, W) \geq p(2p-1)^{n+1}(3p-2)^m = \Phi_{C_{m;\mathbf{k}}}(p).}$$

Moreover,

$$\chi_{C_{m;\mathbf{k}}}(x) = x(x-1)(x-2)^{n+1}(x-3)^m, \qquad \chi(C_{m;\mathbf{k}}) = 4.$$

Thus $[2/3, 1]$ is the complete required interval.

Proof. The chromatic identity follows from $\chi_{K_1 \vee F}(x) = x\chi_F(x-1)$ and Lemma 2.2. Direct substitution into the definition of Φ gives the displayed target.

Theorem 5.1 supplies the base bound for

$$\phi(z) = z^{n+1}(2z-1)^m \qquad (z \in [1/2, 1]).$$

On this interval, ϕ is a product of nonnegative nondecreasing affine factors. Its first and second derivatives are sums of products of nonnegative factors and positive slopes. Hence $\phi, \phi', \phi'' \geq 0$, and $\phi(1/2) = 0$. Apply Lemma 6.1 with $a = 1/2$; its threshold is

$$\frac{1}{2 - 1/2} = \frac{2}{3}.$$

□

7 Thresholds and equality

At the required threshold $p = 1/2$, the target is zero. The proof above is still valid because the only normalized measure divides by p, not by $2p-1$ or by a triangle density. Equivalently, the threshold conclusion follows immediately from nonnegativity.

For every integer $\ell \geq 2$, the balanced complete ℓ-partite graphon T_ℓ has $p = 1 - 1/\ell$ and

$$t(P_{m;\mathbf{k}}, T_\ell) = \frac{\chi_{P_{m;\mathbf{k}}}(\ell)}{\ell^{m+n+2}} = \Phi_{P_{m;\mathbf{k}}}(p).$$

Thus the theorem has the expected Turán equality cases, including T_2 at the threshold. At $p = 1$, both sides are again 1.

For the cone family, the target vanishes at $p = 2/3$, and the balanced complete tripartite graphon attains equality because every cone contains a K_4. More generally, the same colouring-density calculation gives equality on every balanced multipartite grid point $p = 1 - 1/\ell$ with $\ell \geq 3$.

8 Catalogue consequences

Corollary 8.1 (New positive Atlas cases)**.** *The following previously open graphs satisfy $t(H, W) \geq \Phi_H(p)$ for every graphon throughout the full required interval $p \in [1/2, 1]$:*

Atlas	*graph6*	v	e	$\mathbf{k}$	$\chi_H(x)$	$\Phi_H(p)$
41	`Db[`	5	6	$(1,0)$	$x(x-1)^2(x-2)^2$	$p^2(2p-1)^2$
114	`EJwG`	6	7	$(2,0)$	$x(x-1)^3(x-2)^2$	$p^3(2p-1)^2$
138	`Ez‘_`	6	8	$(1,0,0)$	$x(x-1)^2(x-2)^3$	$p^2(2p-1)^3$

For Atlas 41 and 114 one has $m = 2$; for Atlas 138 one has $m = 3$.

Proof. For an unambiguous finite identification, use the canonical labelling from Definition 2.1: the spine is 01, the pages are $2, 3, \ldots, m+1$, and the leaves follow the page vertices.

For Atlas 41 the canonical edge list is

$$01, 02, 03, 12, 13, 24.$$

It is carried to the NetworkX Atlas edge list

$$01, 13, 14, 23, 24, 34$$

by

$$0 \mapsto 3, \quad 1 \mapsto 4, \quad 2 \mapsto 1, \quad 3 \mapsto 2, \quad 4 \mapsto 0.$$

For Atlas 114 the canonical edge list is

$$01, 02, 03, 12, 13, 24, 25.$$

It is carried to

$$04, 12, 13, 14, 23, 24, 45$$

by

$$0 \mapsto 1, \quad 1 \mapsto 2, \quad 2 \mapsto 4, \quad 3 \mapsto 3, \quad 4 \mapsto 0, \quad 5 \mapsto 5.$$

For Atlas 138 the canonical edge list is

$$01, 02, 03, 04, 12, 13, 14, 25.$$

It is carried to

$$01, 02, 04, 12, 13, 15, 23, 25$$

by

$$0 \mapsto 1, \quad 1 \mapsto 2, \quad 2 \mapsto 0, \quad 3 \mapsto 3, \quad 4 \mapsto 5, \quad 5 \mapsto 4.$$

These relabellings verify the structural hypotheses exactly. The orders, sizes, chromatic polynomials, and targets now follow from Lemma 2.2, and Theorem 5.1 proves all three rows. □

Corollary 8.2 (New positive cone: Atlas 193)**.** *NetworkX Atlas graph* 193*, with graph6 code* `En}G`*, satisfies, for every graphon of edge density* $p \in [2/3, 1]$*,*

$$t(F_{193}, W) \geq p(2p-1)^2(3p-2)^2 = \Phi_{F_{193}}(p).$$

It has order 6*, size* 11*, chromatic number* 4*, and*

$$\chi_{F_{193}}(x) = x(x-1)(x-2)^2(x-3)^2.$$

Proof. This graph is the cone over $P_{2;(1,0)}$, the representative used for Atlas 41 above. Add cone vertex 5 to its canonical labelling. The resulting edge list is

$$01, 02, 03, 05, 12, 13, 15, 24, 25, 35, 45.$$

The NetworkX Atlas 193 edge list is

$$01, 03, 04, 05, 12, 13, 14, 23, 24, 34, 45.$$

The first is carried to the second by

$$0 \mapsto 1, \quad 1 \mapsto 3, \quad 2 \mapsto 0, \quad 3 \mapsto 2, \quad 4 \mapsto 5, \quad 5 \mapsto 4.$$

Theorem 6.2 with $(m, \mathbf{k}) = (2, (1, 0))$ now gives the claimed polynomial, target, chromatic number, and graphon bound. □

The same family also recovers Atlas 15, 34, and 92 when $m = 1$, and Atlas 115 when $(m, \mathbf{k}) = (2, (1, 1))$. Those rows were already positive by, respectively, the rooted triangle–tree and edge self-amalgamation theorems. Exact enumeration of all members of the base and cone families with at most six vertices shows that the three rows in Corollary 8.1 and the cone in Corollary 8.2 are the only new catalogue consequences.

9 Accepted inputs, hidden-assumption audit, and limitations

The proof is self-contained at the graphon level. Its only analytic inputs are the elementary inequality $ab \geq a+b-1$, generalized Hölder, Jensen, Tonelli–Fubini, and monotonicity of real powers on $[0,1]$. The weighted rooted-triangle estimate is reproved for every real exponent $h \geq 0$, rather than cited only in its previously used integer form.

No atomlessness, standard-Borel representation, regularity, finite-step approximation, injective-copy interpretation, minimum degree, or pointwise lower bound on d is assumed. All integrands are measurable and bounded. Zero degrees cause no quotient. The only division is by the global edge density p, which is at least $1/2$. At $p = 1$ the argument does not substitute into $1/(1-p)$; that endpoint is handled separately.

The boundary of the theorem is exact: all non-spine vertices must be independent page vertices adjacent to both ends of one common edge, and every added vertex must be a private leaf attached to a page vertex. The proof does not attach leaves to arbitrary vertices of an arbitrary positive graph. It does not cover mixtures of page leaves and spine leaves, edges among page vertices or leaves, shared leaves, longer pendant paths, or books whose pages have more than one internal vertex. Such graphs require additional joint rooted estimates and are not promoted by this result.

A What the Lean formalization actually proves

Corollary 8.1 and Corollary 8.2 are formalized: Atlas 41, 114, 138 and 193 are `verified`. The development is `Methods/Link/PageOp.lean` with the three rows in `Methods/PageBook/`. One difference of substance.

Lemma 4.1 is not used, at any m

The compression is the only place the argument leaves elementary territory, and generalized Hölder does not appear in the development. Every instance the catalogue needs follows instead from the weighted Cauchy–Schwarz inequality $(\int A\eta)^2 \leq (\int A)(\int A\eta^2)$, taken at $A(z) = W(x,z)W(y,z)$ and $\eta(z) = d(z)^{s/2}$, which the project already has:

$$H^2_{(a+b)/2} \leq H_a H_b \qquad (0 \leq a \leq b),$$

that is `sq_pageOp_le'`. For the two-page rows 41 and 114 this is (6) at $m = 2$, one application. For the three-page row 138, where (6) reads $H^3_{1/3} \leq H_0^2 H_1$, two applications suffice:

$$H^4_{1/3} \leq H_0^2 H^2_{2/3} \leq H_0^2 H_{1/3} H_1,$$

from $H^2_{1/3} \leq H_0 H_{2/3}$ and $H^2_{2/3} \leq H_{1/3} H_1$, and cancelling one factor of $H_{1/3}$ (the case $H_{1/3} = 0$ being immediate). This is `cube_pageOp_le`.

Lemma 3.1 is formalized at real exponents, as the note anticipates, and `weighted_rootedTriangle_rpow` sits alongside the $\mathbb{N}$-indexed form that the clique common-leaf and cone families consume unchanged. `Real.rpow` sets $0^0 = 1$, which is the note's convention $d^0 = 1$ including where the degree vanishes.